Boundary Fields in the Spin-One Heisenberg Chain: Physical-Metric Thermal Certificates, Polynomial Moment Filters, and a Sharper Conditional Gap Bootstrap

Dr. Muralidhara Subbarao
Technical Report Assisted by AI Models
www.integralresearch.net
Stony Brook University
Technical Report TR-Ram-OpenAI-722-M544

October 7, 2026

Confidential research report. Private corpus material remains local.

Abstract

We audit the pinned manuscript *A boundary-field gap for the spin-one Heisenberg chain*, emphasizing endpoint orientation, physical metrics in integer coordinates, and the distinction between boundary-weighted spatial moments and physical Gibbs purities. A proved physical-metric Horner bound replaces the source's 193-stage, 2^{50}-scaled thermal schedule by a 107-stage, 2^{49}-scaled schedule with the same certified Frobenius error threshold. Independent bounded integer computations reproduce all fifteen printed thermal quotients and certify all fifteen replacement traces. A rational cubic filter gives a small strictly certified improvement in the normalized leading-overlap bound. More consequentially, retaining the source's level-five seed closes a common quadratic recurrence with coefficient six and yields the conditional uniform gap bound $\log(125000/324)/392$ at the same odd lengths and field. Its periodic spectral premises and four long-chain variational inputs have not been independently established here, so neither the source gap theorem nor its topological consequence is claimed as independently verified. Classical techniques are distinguished from their organization in the private Ram corpus. Contribution assessment: C3/5, a proved sharper conditional boundary bootstrap and independently checked lower-stage thermal certificates give a notable local gain while periodic inputs and long-chain variational premises remain outside independent verification.

Contents

1 Pinned Problem and Exact Mathematical Contract **2**
1.1 Identity, pin, and complete source read 2
1.2 Physical spaces, signs, and quantifiers 2

2 Stepwise Dependency Audit of the Entire Source **3**
2.1 Transfer and endpoint orientation 3
2.2 Periodic inputs remain inputs 4
2.3 Boundary recurrence and initialization 4
2.4 Trial appendix, signed comparison, and topology 5

3 Specific Ram Correspondences and Boundaries **6**

4 Physical-Metric Horner Certificates **7**
4.1 Finite matrices and transported norm 7
4.2 Residual account for a general integer schedule 8
4.3 Review refinement of the unchanged source schedule 9
4.4 Trace intervals retain all errors 9

5 Projected Polynomial Moment Filters **10**
5.1 Positive measure and the appropriate normalization 10
5.2 A small strictly certified improvement 11

6 Sharper Conditional Boundary Bootstrap **11**
6.1 Premises, retained seed, and index alignment 11
6.2 Sharp purity inversion and its limited role 14

7 Executed Verification and Honest Scope **14**
7.1 Author computations actually executed 14
7.2 Independent review and its exact scope 15

8 Comparison and Research Directions **15**

9 Contribution Assessment **16**

1 Pinned Problem and Exact Mathematical Contract

1.1 Identity, pin, and complete source read

This is manuscript M544, not its periodic companion M543. The catalogue title is *A boundary-field gap for the spin-one Heisenberg chain*, source date September 24, 2026, at pin

`adc7f1241b42e322a6451854ab7e4b4c146bf78a.`

The local source directory relative to the study root is

`_sources/openai-math/preprints/A-boundary-field-gap-for-the-spin-one-Heisenber g-chain-September-24-2026.`

The full `build/main.tex`, all nine files in `build/sections`, the bibliography, and both README files were read. This includes every argument and table in `thermal.tex` and `trials.tex`. The local verification directory contains only its README: the three advertised programs and stored JSON certificates are absent. Our thermal replay reconstructs the computation from the printed definitions. Input SHA-256 hashes and read/check scope accompany the author sidecars in Section 7.

1.2 Physical spaces, signs, and quantifiers

For every integer $n \geq 2$, the physical Hilbert space is $\mathcal{H}_n = (\mathbb{C}^3)^{\otimes n}$ with its ordinary tensor-product inner product, conjugate linear in the first slot. Every finite matrix has domain the whole space. In the orthonormal basis (e_{-1}, e_0, e_1),

$$S^z e_s = s e_s, \quad S^+ e_s = \sqrt{2 - s(s+1)}\, e_{s+1} \ (s < 1), \quad S^+ e_1 = 0, \quad S^- = (S^+)^*.$$

Set $S^x = (S^+ + S^-)/2$, $S^y = (S^+ - S^-)/(2i)$, so $\sum_\alpha (S^\alpha)^2 = 2I$, and fix

$$h = \frac{3}{5}, \quad a = \frac{700741}{500000}, \quad b_0 = \frac{49}{4}, \quad b_j = 2^j b_0. \tag{1.1}$$

With unit antiferromagnetic bilinear coupling,

$$H_n^{\mathrm{opp}} = \sum_{r=0}^{n-2} S_r \cdot S_{r+1} - hS_0^z + hS_{n-1}^z, \tag{1.2}$$

$$H_n^{\mathrm{s}} = \sum_{r=0}^{n-2} S_r \cdot S_{r+1} - hS_0^z - hS_{n-1}^z. \tag{1.3}$$

Neither has a closing bond. The periodic operator $H_n^{\mathrm{per}} = \sum_{r=0}^{n-2} S_r \cdot S_{r+1} + S_{n-1} \cdot S_0$ for $n \geq 4$ has no fields. Define full-space unnormalized traces

$$Z_n(b) = \operatorname{Tr} e^{-b(H_n^{\mathrm{per}} + anI)}, \quad Y_n(b) = \operatorname{Tr} e^{-b(H_n^{\mathrm{opp}} + anI)}, \quad Y_n^{\mathrm{s}}(b) = \operatorname{Tr} e^{-b(H_n^{\mathrm{s}} + anI)}.$$

The per-site shift also applies to open chains and cancels in physical purities and energy differences. No magnetization-sector restriction or division by 3^n is present. The pinned headline assertion is

$$E_1(H_{2L+1}^{\mathrm{s}}) - E_0(H_{2L+1}^{\mathrm{s}}) > \frac{\log 10}{392} \quad \text{for every integer } L \geq 960, \tag{1.4}$$

with eigenvalues counted with multiplicity. This report does not promote that assertion to an independently verified theorem. The stronger bound below is conditional. Neither is asserted here for free ends, all $h > 0$, arbitrary perturbations, or every infinite-chain ground state.

2 Stepwise Dependency Audit of the Entire Source

2.1 Transfer and endpoint orientation

For fixed $b > 0$, let Ω_b be the disjoint union of simplexes $0 < t_1 < \cdots < t_d < b$ marked by $\alpha_r \in \{x, y, z\}$, including the empty history. Lebesgue measure on each simplex and unit mass at the empty history define μ_b, with total mass e^{3b}. The auxiliary space is $\mathcal{K}_b = L^2(\Omega_b, \mu_b; \mathbb{C})$. In Cartesian spin coordinates, $G_\alpha = iS^\alpha$ is real, $\|G_\alpha\| = 1$, and $-S \cdot S = \sum_\alpha G_\alpha \otimes G_\alpha$. Chronologically merge two histories and trace their matrix product to obtain $k_b(\omega, \eta)$, zero at nonempty collisions and 3 for two empty histories. The source defines

$$(X_b f)(\omega) = e^{-ab} \int k_b(\omega, \eta) f(\eta)\, d\mu_b(\eta). \tag{2.1}$$

Symmetry follows by exchanging histories, which preserves their merged list, not by reversing noncommuting products. Since $|k_b| \leq 3$ and the measure is finite, X_b is bounded, self-adjoint, and Hilbert–Schmidt, with domain all of $\mathcal{K}_b$. Its nonzero eigenvalues are square summable; powers of degree at least two are trace class.

The group $\mathcal{G}$ of twelve proper signed coordinate permutations acts by signed history relabeling via real unitaries U_g commuting with X_b. The boundary vector is $v_b(\omega) = (ih)^d$ when all marks are z, zero otherwise, with empty value one. Its norm is $\|v_b\|^2 = e^{bh^2}$, not one. For $P = \operatorname{diag}(1, -1, -1)$ the source's open-chain identity is

$$Y_n(b) = \langle v_b, X_b^n v_b \rangle, \qquad Y_n^{\mathrm{s}}(b) = \langle v_b, X_b^n U_P v_b \rangle. \tag{2.2}$$

The conjugated left coefficient gives $(-ih)G_z = hS^z$ in the negative Hamiltonian; the right coefficient gives $(ih)G_z = -hS^z$. Applying P flips only the right field. Regrouping the time expansion is justified by the absolute bound $3^n \exp(b[3(n-1)+2h])$ before shifting. There are n kernels and $n-1$ bonds. This is not unitary equivalence of the two physical Hamiltonians. Even opposite-field moments are positive moments of $|X_b|$; odd powers retain spectral signs.

2.2 Periodic inputs remain inputs

Source `periodic.tex`, labels `per:native` and `per:concentration`, imports the companion identity $Z_n(b) = \operatorname{Tr} X_b^n$ and finite data. With

$$\ell = .999, \quad U = 1.00139, \quad V = .872, \quad m_J = 1.0657205, \quad m_N = .2783155,$$

the D_2-fixed sector has twelfth moment at most m_J, spectral cap U, and an eigenvalue of modulus above ℓ. Three other sectors each have twelfth moment at most m_N and cap V. Additional inputs are $1 - S(72, b_1) \le .0604996$ and negative shifted periodic ground energies at lengths $72, 120$. Our open-chain thermal tests do not establish these. The distinct periodic purities are

$$S(n,b) = Z_{2n}(b)/Z_n(b)^2, \qquad T(n,b) = Z_n(2b)/Z_n(b)^2,$$

for spatial eigenvalue weights and physical Gibbs weights, respectively. The source couples them by exact cancellation and

$$f(u) = \frac{u^2}{2(1-u)^2}, \qquad d(u) = \frac{u}{2-3u} \quad (0 \le u \le .1). \tag{2.3}$$

Normalized squaring sends a purity defect at most u to one at most $f(u)$; concentration gives residual-to-largest ratio at most $d(u)$. The source's simple leading spatial eigenvalue is positive by the high odd-trace identity and positivity of physical partition functions. Its real line is group-fixed: the order-three generator acts by $+1$, and its three conjugate half-turn signs equal their own square. For $j \ge 2$ the needed quantitative outputs are

$$\begin{aligned} &n_j = 120\, 2^{j-2}, \quad M_j = 2n_j, \quad u_2 = .052, \quad u_{j+1} = 6u_j^2, \\ &\rho_j^{n_j} \le d(u_j), \qquad 0 < r_j := \lambda_{j+1}/\lambda_j^2 \le 1, \qquad r_j^{M_j} \ge (1-u_j)(1-u_{j+1}), \end{aligned} \tag{2.4}$$

where $\lambda_j = \|X_{b_j}\|$ and ρ_j is the largest residual modulus divided by λ_j. Initially $\rho_0 < .874$, $\rho_1 < .9525$, $\rho_2 < .971$. An abstract periodic gap alone does not supply these specific bounds.

2.3 Boundary recurrence and initialization

Let P_j project onto the leading line. For positive even k, m set

$$\begin{aligned} c_j &= \|P_j v_{b_j}\|^2, \quad B_k(j) = \langle v_{b_j}, (|X_{b_j}|/\lambda_j)^k v_{b_j} \rangle, \\ e_k(j) &= (B_k(j) - c_j)/B_k(j), \qquad A(m,j) = Y_m(b_{j+1})/Y_m(b_j)^2. \end{aligned} \tag{2.5}$$

Here $B_k > 0$, $0 \le e_k \le 1$, and A is a physical purity in $(0,1]$. The endpoint weights change with temperature. At fixed temperature,

$$0 \le B_m(j) - c_j \le \rho_j^{m-k}(B_k(j) - c_j) \quad (m \ge k). \tag{2.6}$$

No changing-boundary-vector norm enters. For $e_k(j) < 1$ and $0 < q < 1$ with $q \ge \rho_{j+1}^{m-k}$, the source proves

$$e_m(j+1) \le F(q, e_k(j), A(m,j)), \qquad F(q,e,t) = \frac{q}{1-q}\left(\frac{1}{(1-e)^2 t} - 1\right). \tag{2.7}$$

The new-temperature contraction gives the bound $q(B_k(j+1)/B_m(j+1) - 1)/(1-q)$. Exactly,

$$\frac{B_k(j+1)}{B_m(j+1)} = \frac{A(k,j)}{A(m,j)}\left(\frac{B_k(j)}{B_m(j)}\right)^2 r_j^{m-k}.$$

Use $A(k,j) \le 1$, $r_j \le 1$, $B_m(j) \ge c_j$. Only the old overlap is divided by; a bound $e_m(j+1) < 1$ proves the new overlap positive without assuming it.

Set $k_2 = 142$, $k_{j+1} = M_j$. Bounds $e_{k_j}(j) \le E_j$ and $1 - A(M_j, j) \le \eta_j$ propagate if both next bounds are below one and

$$\begin{aligned} E_{j+1} &\ge F\left(d(u_{j+1})^{(M_j - k_j)/M_j}, E_j, 1-\eta_j\right), \\ \eta_{j+1} &\ge f\left(1 - (1-\eta_j)(1-u_j)(1-u_{j+1})(1-E_{j+1})\right). \end{aligned} \tag{2.8}$$

The physical update follows, for $M = M_j$, from

$$\frac{A(2M,j)}{A(M,j)} = r_j^M \frac{B_{2M}(j+1)}{B_M(j+1)}\left(\frac{B_M(j)}{B_{2M}(j)}\right)^2$$

and physical Gibbs squaring at fixed length $2M$.

Source `initialization.tex` reaches $E_2 = .14, \eta_2 = .21$ using $Y_6(b_0) < .027$, $c_0 P_*(\lambda_0)^2 < .003745$, $P_*(x) = x^3 - .14x^2 - .18x$, and

$$E_0(H_m^{\mathrm{opp}}) + am < \alpha := .4042 \quad (m = 34, 78, 142, 240). \tag{2.9}$$

The intermediate quantities are

$$\epsilon(m) = .027 e^{b_0 \alpha} U^m .874^{m-6}/\ell^6, \qquad R(m) = e^{.132} U^m + \epsilon(m) - 1.$$

They give $e_m(0) \le \epsilon(m)$ and $A(m,j) \ge (1 + R(m)^{2^j})^{-2}$. The sixth moment bounds residual mass, the cubic bounds the leading term, and trial energies bound physical ground weights. Two steps use lengths $(34, 78)$ and $(78, 142)$, before reference length 240.

2.4 Trial appendix, signed comparison, and topology

The trial appendix specifies integer 26×26 matrices A_s on virtual spin types $(1,1,1,1,3,3,3,5)$, with selection rule $e = d + 2s$. Both endpoints have virtual weight -1. Physical amplitudes $\mathbf{b}^t A_{s_1} \cdots A_{s_m} \mathbf{c}$ are ordinary real coordinates; no virtual Gram factor belongs in their physical norm. The norm transfer $T_* = \sum_s A_s \otimes A_s$, field insertion $G_* = \sum_s s A_s \otimes A_s$, and bond insertion O_* preserve weight difference. Multiplicities $1, 4, 8, 8, 4, 1$ give an exact invariant 162-dimensional block, not an uncontrolled compression. With $L_i = L_0 T_*^i$,

$$N_1 = -300000 L_0 G_*, \qquad N_i = N_{i-1} T_* + L_{i-2}(500000 O_* + 700741 T_*^2).$$

The readout $V_m = L_m u$ and $H_m^* = (N_m + 300000 L_{m-1} G_* + 700741 L_m)u$ has $m-1$ bonds, negative left and positive right fields, and m shifts. The printed $V_m > 0$ and $202100 V_m - H_m^* > 0$ imply (2.9). All tables and the contraction algebra were read; the four long integer contractions were not replayed.

For even M and every integer $n \geq M$, the fixed leading line and Cauchy–Schwarz on each residual eigenspace give

$$\left|Y_n^{\mathrm{s}}(b_j)/\lambda_j^n - c_j\right| \leq B_M(j) - c_j. \tag{2.10}$$

This includes negative eigenvalues and odd physical lengths. The source then uses common even reference lengths at consecutive temperatures to control physical purity and p_1/p_0.

The topology section concerns every subsequential local limit, not full-sequence convergence or uniqueness among all infinite-volume ground states. A uniform gap gives, for every local observable A,

$$\omega(A^*[H_{\mathrm{bulk}}, A]) \geq \gamma(\omega(A^*A) - |\omega(A)|^2).$$

Endpoint terms eventually commute with each fixed A. The twist is

$$U_\varepsilon = \exp\left[-i \sum_{j \in \mathbb{Z} \cap [-\pi/\varepsilon, \pi/\varepsilon]} (\pi + \varepsilon j) S_j^z\right].$$

Tasaki's Assumption 2 and Theorem 3 concern this odd open same-field setting, not a periodic-only gap [3]. The limit order is first a subsequential local limit at fixed ε, then $\varepsilon \downarrow 0$. Index -1 remains conditional on the required gap here. The extra global uniqueness needed for Tasaki's half-chain corollary is not proved.

3 Specific Ram Correspondences and Boundaries

The following actual local passages were read, not merely routing maps. Absolute paths identify corpus revisions.

1. `C:/Users/mural/OneDrive/Documents/RamCalBooks/volumes/volume21/chapters/ch01d-physical-metrics.tex`, *A Nonorthogonal Representation Carries a Metric*, labels `v21c01:physical-gram-definition` and `v21c01:metric-symmetry`: the physical norm follows synthesis. For M544 coordinates synthesis is J^{-1}, not J, so the Gram matrix is J^{-2}. *The Full Residual Has an Omitted Component*, `v21c01:residual-decomposition` and `v21c01:residual-identity`, distinguishes compressed and full residuals. *Approximating the Metric*, `v21c01:gram-equivalence`, requires relative form bounds if this exact metric is later approximated.

2. `C:/Users/mural/OneDrive/Documents/RamCalBooks/volumes/volume21/chapters/ch01a-state-domain.tex`, *Transporting a Closed Realization*, `v21c01:bounded-similarity`: coordinates transport domain and norm. Here the matrices are finite and everywhere defined, and $1 \leq J_{ww} \leq \sqrt{2}$; no unbounded-transform claim is needed.

3. `C:/Users/mural/OneDrive/Documents/RamCalBooks/volumes/volume21/chapters/ch02/finite.tex`, *The polynomial functional*, `v21c02:hankel` and `v21c02:localizer`, identifies a quadratic form with a square integral. This matches the projected boundary measure below. Labels `v21c02:zero-energy` and `v21c02:atomic-certificate` require exact null polynomials and complete checks for atomic recovery. A small positive filter value is neither an annihilator nor a finite-spectrum theorem.

4. `C:/Users/mural/OneDrive/Documents/RamPatsCore50-TRs/TR-core-pat-5/TR-core-pat-5.tex`, *Polynomial evaluation without eigendecomposition*, `eq:filtererror`, transfers scalar error to self-adjoint operators. M544 requires its physical metric for this step. *Residuals*

of an inexact Schur reconstruction, `prop:schurresidual/eq:schurresidual`, motivates a full residual account if elimination is introduced. The *Low-rank update identity*, `thm:woodbury`, is not used: an endpoint operator is not automatically low rank on the full tensor-product space.

These passages organize classical metric, spectral, residual, and moment facts into explicit problem and acceptance records. We use that organization for source-specific certificates, without attributing the spectral theorem, Horner rule, Schur algebra, Poisson truncation, or Christoffel minimization to historically new Ram discoveries. The source already handles its similarity metric correctly; this report refines its quantitative certificate, not an alleged missing metric.

4 Physical-Metric Horner Certificates

4.1 Finite matrices and transported norm

For $2 \le n \le 6$ and $g \in \{1, -1, 0\}$ define

$$H_{n,g} = \sum_{r=0}^{n-2} S_r \cdot S_{r+1} - \tfrac{3}{5} S_0^z + \tfrac{3}{5} R_g, \qquad R_{\pm 1} = \pm S_{n-1}^z, \quad R_0 = S_{n-1}^x.$$

Write $Z_{n,g} = \operatorname{Tr} e^{-b_0(H_{n,g}+anI)}$ and

$$C_n = 3\left\lfloor \frac{n-1}{2} \right\rfloor + 2((n-1) \bmod 2) + \frac{6}{5}, \qquad A = I - \frac{H_{n,g} + C_n I}{8}. \tag{4.1}$$

A bond has eigenvalues $-2, -1, 1$; two consecutive bonds have minimum -3, by coupling the outer spins and then the middle spin. Adding these bounds and the field norm $6/5$ gives $0 \le H_{n,g} + C_n I \le 16I$. Thus $A = A^*$ and $-I \le A \le I$; positivity of A is unnecessary. For words $w \in \{-1, 0, 1\}^n$, let $p_w = (\sum_r w_r) \bmod 2$ and

$$J_{ww} = \sqrt{1 + p_w}, \quad D = 80JAJ^{-1}, \quad G = J^{-2} = \operatorname{diag}((1 + p_w)^{-1}).$$

The integer diagonal is $80 - 10C_n - 10\sum_r w_r w_{r+1} + 6w_0 - 6gw_{n-1}$. Allowed bond hops have entry -10; for $g = 0$ an allowed end hop from column r to row w has entry $-3(1 + p_w)$. Nonsymmetry is required.

Proposition 4.1 (Metric matrix norm)**.** *For every matrix B on these coordinates,*

$$\|B\|_{G,F}^2 := \operatorname{Tr}(G^{-1}B^*GB) = \|J^{-1}BJ\|_F^2 = \sum_{w,r} \frac{1 + p_r}{1 + p_w} |B_{wr}|^2. \tag{4.2}$$

Moreover $D^*G = GD$ and $\|(D/80)B\|_{G,F} \le \|B\|_{G,F}$.

Proof. Substitute $D = 80JAJ^{-1}$, $G = J^{-2}$, $A = A^*$. Cyclicity of trace proves the norm identity. Conjugating left multiplication gives A, a Frobenius contraction because $\|A\| \le 1$. □

Euclidean contraction of $D/80$ is not asserted. Similar spectra alone would not justify that conclusion.

4.2 Residual account for a general integer schedule

For $K \geq 49$ and a positive integer Q, define

$$q_i = \frac{49^i/i!}{\sum_{r=0}^{K} 49^r/r!}, \quad t_i = \lfloor Q q_i \rfloor, \quad \mathcal{X}_{K+1} = 0, \quad \mathcal{X}_i = \left\lfloor \frac{D\mathcal{X}_{i+1}}{80} \right\rfloor + t_i I. \tag{4.3}$$

Floors are entrywise toward $-\infty$, including negative arguments; every stage uses the complete old matrix. Put

$$\widehat{E} = J^{-1}(\mathcal{X}_0/Q)J, \quad E = e^{-49}e^{49A} = e^{-b_0(H_{n,g}+C_n I)/2}, \quad R_i = \frac{\mathcal{X}_i}{Q} - \frac{D}{80}\frac{\mathcal{X}_{i+1}}{Q} - q_i I.$$

Theorem 4.2 (Horner residual certificate)**.** *Let $d = 3^n$ and*

$$\tau_K = \frac{49^{K+1}}{(K+1)!}\frac{1}{1-49/(K+2)}\left(\sum_{r=0}^{K}\frac{49^r}{r!}\right)^{-1}. \tag{4.4}$$

Then

$$\widehat{E} - \sum_{i=0}^{K} q_i A^i = \sum_{i=0}^{K} A^i J^{-1} R_i J,$$

$$\|\widehat{E} - E\|_F \leq 2\sqrt{d}\,\tau_K + \sum_{i=0}^{K}\|R_i\|_{G,F}. \tag{4.5}$$

The computed $\widehat{E}$ need not be self-adjoint.

Proof. Subtract exact polynomial Horner recursion after conjugating the computed recursion and unroll. Contraction by A^i gives the residual sum. Omitted exponential terms have consecutive ratio at most $49/(K+2) < 1$, bounding their sum by the first two factors in (4.4). Dividing by e^{49} is bounded by division by its partial sum. The omitted Poisson mass is thus at most τ_K. Renormalizing retained coefficients changes the full coefficient list by twice the omitted mass in ℓ^1, the classical normalized-tail principle [6]. Now use $\|A^i\|_F \leq \sqrt{d}$. □

This permits a posteriori rational residual certificates. The executed schedule instead uses the following explicit a priori bound.

Corollary 4.3 (Parity-resolved rounding budget)**.** *Let a_n, b_n count even and odd total-parity words. Then*

$$\|\widehat{E} - E\|_F < 2\sqrt{d}\,\tau_K + \frac{K+1}{Q}\left(\sqrt{d} + \sqrt{(a_n + b_n/2)(a_n + 2b_n)}\right), \tag{4.6}$$

where $a_n = (3^n + (-1)^n)/2$, $b_n = (3^n - (-1)^n)/2$.

Proof. Coefficient floors contribute less than $\sqrt{d}/Q$ per stage because similarity fixes I. Matrix floor residual entries have modulus below $1/Q$; summing their squared weights in (4.2) gives the product displayed. The signed parity word count is $(1-1-1)^n = (-1)^n$, and the total is 3^n. Sum over all stages. □

At $n = 6$ the squared rounding factor is $547 \cdot 1093 = 597871 < 774^2$; smaller n have no larger factor and $\sqrt{d} \leq 27$.

Theorem 4.4 (A 107-stage thermal replacement)**.** *For every* $2 \le n \le 6$ *and* $g \in \{1, -1, 0\}$*, taking* $K = 106$*,* $Q = 2^{49}$ *gives*

$$\|\widehat{E} - E\|_F < 1.83 \cdot 10^{-10} < 2 \cdot 10^{-10},$$

the source's same final error threshold.

Proof. Exact rational comparison gives $\tau_{106} < 556 \cdot 10^{-15}$ and

$$54\tau_{106} + \frac{107 \cdot 801}{2^{49}} < 183 \cdot 10^{-12}.$$

Apply Corollary 4.3. Both comparisons were executed. □

This changes the nominal schedule from 193 to 107 stages and halves Q. It is not a measured runtime speedup. The review refinement below shows why nominal stage counts substantially overstate the relevant comparison with a source implementation that skips zero coefficients.

4.3 Review refinement of the unchanged source schedule

Independent review found, by exact scalar arithmetic for $K = 192, Q = 2^{50}$, that $t_i = 0$ for $i > 114$, $t_{114} > 0$, and $Q - \sum_i t_i = 55$. Starting with zero, stages $192, \ldots, 115$ are inert, and the matrix product at stage 114 also has zero input. Thus at most 114 nonzero-input floor operations remain, not 193.

For any integer numerator divided by 80, the floor residual has modulus at most 79/80, even for negative numerators. The exact full squared weight count is

$$(a_n + b_n/2)(a_n + 2b_n) = (9d^2 - 1)/8.$$

For $g = \pm 1$, total magnetization is preserved by D, the identity, and every entrywise floor; off-block entries remain exactly zero. Within each block J is scalar. The squared count is then $\sum_s d_s^2 = [z^0](z^{-1} + 1 + z)^{2n}$, where d_s is a magnetization multiplicity, and equals 73789 at $n = 6$. With the source's unnormalized tail $R = 49^{193}/193!/(1 - 49/194) < 3.152 \cdot 10^{-33}$, one consequently has

$$\|\widehat{E} - E\|_F \le 2\sqrt{d}\, R + \frac{55\sqrt{d} + (79/80)\, 114\sqrt{W_{\text{floor}}}}{2^{50}}. \tag{4.7}$$

To prove this, sum coefficient deficits exactly (each is nonnegative), use the source's Poisson-tail bound, and apply contractive propagation to the 114 potentially nonzero physical floor residuals. The review's rational square-root checks give bounds below $7.864 \cdot 10^{-11}$ for $g = 0$ and $2.848 \cdot 10^{-11}$ for $g = \pm 1$. This sharpens the unchanged certificate without altering any quotient. The fair implementation comparison is thus 115 retained Horner stages after removing inert leading zeros, with 114 potentially nonzero-input matrix products, versus at most 107 retained stages in the replacement, not a claimed 44.56% runtime saving. Some low-index coefficients also vanish; a retained stage need not have a nonzero coefficient.

4.4 Trace intervals retain all errors

For $W = \|\widehat{E}\|_F^2$, the integer floor is

$$k = \lfloor 10^{16} W \rfloor = \left\lfloor \frac{10^{16}}{2Q^2} \sum_{w,r} \mathcal{X}_0(w, r)^2 (2 - p_w)(1 + p_r) \right\rfloor. \tag{4.8}$$

Set $L_k = k/10^{16}$, $U_k = (k+1)/10^{16}$, $r_k = (\lfloor\sqrt{k+1}\rfloor + 1)/10^8$, and $\varepsilon_k = 4 \cdot 10^{-10} r_k + 4 \cdot 10^{-20}$. All executed cases have $L_k > \varepsilon_k$, and

$$e^{b_0(C_n - an)}(L_k - \varepsilon_k) < Z_{n,g} < e^{b_0(C_n - an)}(U_k + \varepsilon_k). \tag{4.9}$$

Indeed the squared-norm error is at most $2\|\widehat{E}\|_F\|E - \widehat{E}\|_F + \|E - \widehat{E}\|_F^2$. For $0 \le t \le 10$, use $T_{80}(t) \le e^t \le T_{80}(t) + 2t^{81}/81!$, where $T_{80}(t) = \sum_{r=0}^{80} t^r/r!$, and reciprocal intervals for negative arguments. The arguments lie in $[-.0544635, 9.691073]$. Thus floors, coefficients, tail, squaring, and rescaling are all covered by rational trace enclosures.

5 Projected Polynomial Moment Filters

5.1 Positive measure and the appropriate normalization

Let $\Pi = |\mathcal{G}|^{-1} \sum_g U_g$ and define the positive finite spectral measure

$$\nu(B) = \langle \Pi v_{b_0}, \mathbf{1}_B(X_{b_0}) \Pi v_{b_0} \rangle.$$

Under the periodic premises its support is in $[-U, U]$, and its mass is $\|\Pi v_{b_0}\|^2 \le e^{b_0 h^2}$, not one. The orbit of z consists of the six signed axes with equal multiplicity; global rotations around the left z axis identify the four perpendicular right fields. Hence for $n \ge 2$,

$$I_n := \int x^n \, d\nu(x) = \frac{Z_{n,1} + Z_{n,-1} + 4Z_{n,0}}{6}. \tag{5.1}$$

If the leading line is group-fixed, $\nu(\{\lambda_0\}) = c_0$. A positive measure does not make arbitrary odd moments positive. Here the specific partition-function representation does, but its sign is unnecessary for the square-polynomial bound. With $\phi(x) = (x, x^2, x^3)^t$ and $M = (I_{i+j})_{i,j=1}^3$,

$$c_0 |c^t \phi(\lambda_0)|^2 \le \int |c^t \phi(x)|^2 \, d\nu(x) = c^t M c \quad (c \in \mathbb{R}^3). \tag{5.2}$$

This Hankel matrix corresponds to $x^2 d\nu$, using only $I_2, \dots, I_6$. For P_* the expression is $I_6 - .28 I_5 - .3404 I_4 + .0504 I_3 + .0324 I_2$, exactly as in the source.

Proposition 5.1 (Normalized filter minimization)**.** *If $M \succ 0$ and $u > 0$, the least $c^t M c$ subject to $c^t \phi(u) = 1$ is $1/(\phi(u)^t M^{-1} \phi(u))$, attained at $c = M^{-1}\phi(u)/(\phi(u)^t M^{-1}\phi(u))$. Independently of invertibility, if $I_k \in [l_k, h_k]$ and $p(x)^2 = \sum_{k=2}^6 a_k x^k$, then*

$$\int p^2 \, d\nu \le \sum_{a_k \ge 0} a_k h_k + \sum_{a_k < 0} a_k l_k. \tag{5.3}$$

Proof. Apply Cauchy–Schwarz to $M^{1/2}c$ and $M^{-1/2}\phi(u)$. The interval assertion is a linear extremum over the enclosing box; it does not assume every point of that box is realizable. □

The minimization is classical Christoffel-type algebra [7]. A midpoint floating minimizer only proposes a polynomial; certification uses rational coefficients and intervals.

5.2 A small strictly certified improvement

Choose

$$p(x) = x^3 - .14241x^2 - .17425x. \tag{5.4}$$

The executed rational intervals prove

$$\begin{aligned} .0080122179 < \frac{\int p^2\, d\nu}{p(U)^2} < .0080122385, \\ .0080133748 < \frac{\int P_*^2\, d\nu}{P_*(U)^2} < .0080133957. \end{aligned} \tag{5.5}$$

These are outward shortened displays of rational bounds; the new upper bound is strictly below the old lower bound. The relative improvement is about 0.014%, not large. Unnormalized comparison is misleading: the new $\int p^2 d\nu$ is approximately .00378017, larger than the old .00374403, because $p(U)$ is also larger.

Proposition 5.2 (Uniform leading contribution)**.** *Assume $\ell < \lambda_0 \le U$ and a group-fixed leading line. For every integer $m > 8$,*

$$e^{b_0\alpha}\lambda_0^m c_0 < 1.133\, U^m. \tag{5.6}$$

Proof. On $[\ell, U]$, $p = x(x^2 - .14241x - .17425) > 0$. Also $4p - xp' = x(x^2 - .28482x - .52275) > 0$, since the quadratic is positive at ℓ and increasing throughout the interval. Thus $xp'/p < 4$, and $\frac{d}{dx}\log(x^m/p(x)^2) > (m-8)/x > 0$. Use (5.2) and monotonicity to get

$$e^{b_0\alpha}\lambda_0^m c_0 \le U^m e^{b_0\alpha}\frac{\int p^2 d\nu}{p(U)^2}.$$

The last coefficient is rigorously below $1.1327688 < 1.133$ by rational exponential enclosure. □

The source's pre-rounding coefficient is below 1.133226, before enlargement to $e^{.132}$. This is a modest local improvement. The stronger bootstrap below uses the original seeds, so its gap improvement does not depend on the alternate cubic or thermal schedule.

6 Sharper Conditional Boundary Bootstrap

6.1 Premises, retained seed, and index alignment

We assume the source's transfer identities, simple positive group-fixed leading lines, bounds (2.4), and initialization $E_2 = .14, \eta_2 = .21$. The last follows from the source's thermal and four trial premises and its finite initialization comparisons; the long-chain trial contractions remain unreplayed. An unqualified periodic gap theorem is not an equivalent premise.

The source's rounded boundary table is

j	E_j	η_j	next-step q upper bound
2	.14	.21	.142
3	.12	.155	.0282
4	.016	.026	.0028
5	.001	.001	—

The author's scalar check verifies the three transitions exactly, including $d(u_3)^{49} < .142^{120}$ and the later squared-root comparisons. Independent review reproduced these outputs:

$$e_{k_5}(5) \le \frac{3112739}{18367982739} < .000170,$$
$$1 - A(M_5, 5) < .000432, \qquad u_5 = \frac{1784003086827}{1192092895507812 50} < .000015. \tag{6.1}$$

For exact reproducibility the physical-defect rational upper output is

$$\frac{689222018199025743319235325507176688589603936578526044964609}{15987970410235323323732105160295177430337229395626101770899292 18}.$$

This output already used the conservative $E_5 = .001$ in its product. It therefore remains a valid bound for the actual physical defect when we separately retain the sharper actual spatial defect. No retroactive change to the previous step or circular use of a new seed is needed. Thus all three actual defects are below $s = .000432 = 27/62500$.

The indices have different roles:

Symbol	Role for $j \ge 5$
j	temperature-doubling index
$b_j = 392\,2^{j-5}$	physical inverse temperature
$n_j = 120\,2^{j-2}$	periodic spatial reference exponent
$M_j = 240\,2^{j-2}$	even boundary physical reference length
$k_j = M_j/2$	old-temperature residual reference length
$k_{j+1} = M_j = n_{j+1}$	common reference at the next temperature
$n = 2L + 1$	actual same-field physical chain length

In particular $M_5 = 1920$, $k_5 = 960$, and the physical chain length n below need not equal any doubling length.

Theorem 6.1 (Seed-preserving common recurrence)**.** *Under these premises let $y_5 = s = .000432$ and $y_{j+1} = 6y_j^2$. For every $j \ge 5$,*

$$u_j \le y_j, \qquad e_{k_j}(j) \le y_j, \qquad 1 - A(M_j, j) \le y_j, \qquad y_j = \frac{1}{6}\left(\frac{324}{125000}\right)^{2^{j-5}}. \tag{6.2}$$

Proof. The base case is (6.1). The following argument works for every common seed $s \le .001$. Suppose the bounds hold at j, write $y = y_j \le .001$, and note $k_j = M_j/2$ and $n_{j+1} = M_j$. By the periodic recurrence and monotonicity of d,

$$q \le \sqrt{d(6y^2)} = y\sqrt{\frac{6}{2 - 18y^2}} \le \frac{7}{4}y.$$

Squaring at .001 proves the last inequality, and its left quotient increases on the entire interval. The function F increases in q, e and decreases in t, so

$$\frac{F(7y/4, y, 1-y)}{y^2} = \frac{7/4}{1 - 7y/4}\sum_{r=1}^{3}(1-y)^{-r}.$$

All positive factors increase with y. At $y = .001$ this is $20979007000000/3981032975007 < 6$. Thus $E_{j+1} = 6y^2 < 1$ is admissible in (2.8), also proving the new leading overlap positive.

The defect of a product of factors in $[0,1]$ is at most the sum of their defects. The physical update product therefore has defect at most $2y+12y^2$. Consequently

$$\frac{f(2y+12y^2)}{y^2}=\frac{(2+12y)^2}{2(1-2y-12y^2)^2}\le\frac{126504500000}{62248753009}<6.$$

The numerator increases and the positive denominator decreases on $[0,.001]$, which justifies the endpoint test for every y. We may set $\eta_{j+1}=6y^2$, and $u_{j+1}\le 6y^2$ closes the third bound. All factors and denominators are positive and $6y^2<y\le .001$, so the interval is invariant. Finally $6y_{j+1}=(6y_j)^2$ gives the explicit formula. □

Corollary 6.2 (Improved conditional gap at every required odd length)**.** *Under the hypotheses of Theorem 6.1,*

$$E_1(H^{\mathrm{s}}_{2L+1})-E_0(H^{\mathrm{s}}_{2L+1})>\frac{\log(125000/324)}{392}>.01519\quad\textit{for every integer } L\ge 960. \tag{6.3}$$

Proof. Fix $j\ge 5$ and any integer $M_j\le n\le 2M_j$. Set $M=M_j$ and $Q_j(n)=Y^{\mathrm{s}}_n(b_j)/\lambda_j^n$. At the old temperature, (2.10) gives $0<Q_j(n)\le B_M(j)$; positivity makes squaring this upper bound legitimate. At the new temperature use the same even $M=k_{j+1}$:

$$Q_{j+1}(n)\ge 2c_{j+1}-B_M(j+1)\ge(1-2y_{j+1})B_M(j+1)>0.$$

Cancelling factors, with the same actual physical length n at both temperatures, gives

$$\begin{aligned}\frac{Y^{\mathrm{s}}_n(b_{j+1})}{Y^{\mathrm{s}}_n(b_j)^2}&\ge(1-2y_{j+1})A(M,j)r_j^{n-M}\\&\ge(1-2y_{j+1})(1-y_j)(1-u_j)(1-u_{j+1})\\&\ge 1-(2y_j+18y_j^2)>1-3y_j.\end{aligned}$$

For the middle inequality, $0<r_j\le 1$ and $0\le n-M\le M$ imply $r_j^{n-M}\ge r_j^M$. This is the step covering every intermediate length. The left side is precisely the physical same-field Gibbs purity at inverse temperature b_j, since $b_{j+1}=2b_j$. List its Gibbs probabilities in decreasing order, with multiplicities. Then $\sum_i p_i^2\le p_0$ implies $p_0>1-3y_j>1/2$, so the physical ground state is simple and

$$e^{-b_j(E_1-E_0)}=\frac{p_1}{p_0}\le\frac{1-p_0}{p_0}<\frac{3y_j}{1-3y_j}<6y_j=\left(\frac{324}{125000}\right)^{2^{j-5}}.$$

Divide the logarithm by $b_j=392\,2^{j-5}$. Finally $M_5=1920$ and $M_{j+1}=2M_j$ show that the closed intervals $[M_j,2M_j]$ cover every integer $n\ge 1920$. All odd $n=2L+1$ with $L\ge 960$ are covered, not just a subsequence. The rational comparison $e^{392(.01519)}<125000/324$ was independently checked with the rational Taylor upper enclosure, proving the last bound. □

The logarithmic constant is approximately .0151921569, compared with the source's .0058739416. These decimals are explanatory, not proof premises. The earlier author seed .001 gives $\log(500/3)/392$ by the same proof. Review sharpened the already available seed; it introduced no new thermal or trial assumption. This is a stronger implication at the actual weighted-boundary step, not an independent proof of all its upstream premises.

6.2 Sharp purity inversion and its limited role

Lemma 6.3 (Binary purity inverse)**.** *For a finite or countable probability list of purity at least* $1-\delta$*,* $0 \le \delta < 1/2$*, the largest weight is unique and*

$$\frac{1-p_0}{p_0} \le \frac{1-\sqrt{1-2\delta}}{1+\sqrt{1-2\delta}} =: R_{\mathrm{bin}}(\delta). \tag{6.4}$$

The bound is attained by a two-point list.

Proof. Purity is at most p_0, so $p_0 \ge 1-\delta > 1/2$. Writing $x = 1-p_0 < 1/2$ and using $\sum_{i>0} p_i^2 \le x^2$ gives $2x(1-x) \le \delta$. Solve on $[0, 1/2)$ and apply the increasing function $x/(1-x)$. Two-point lists saturate both inequalities. □

The hypotheses hold here with $\delta_j = 2y_j + 18y_j^2 < 1/2$. Thus each dyadic physical-length interval also has the lower bound $-b_j^{-1} \log R_{\mathrm{bin}}(\delta_j)$. This classical refinement changes a finite-level prefactor; it alone does not change the limiting exponential rate. The uniform improvement (6.3) instead comes from the retained seed and coefficient-six induction.

7 Executed Verification and Honest Scope

7.1 Author computations actually executed

`_audit/M544-author-check.py` reconstructs all fifteen integer matrices from the printed rules, runs both schedules, and writes `_audit/M544-author-check.json`. All thirty bounded recurrences terminated and passed. Every source quotient matched its printed value exactly. The replacement quotients are

$n\backslash g$	1	−1	0
2	10185296075302	4160017987	583578389575
3	259254948724	571111660441436	47796720049345
4	318278534893	661347985	43971049724
5	260908126899	29917222640308	5659378873686
6	16467340956	328219854	4086161507

These approximate the same exact thermal operators but need not equal the original quotients. The certified intervals prove $Z_{6,1} < .027$ and the original cubic bound below .003745, together with (5.5). Each I_n is within 10^{-7} of the source center.

The program uses signed 64-bit NumPy arrays only after checking the Python-integer bound $\max_w \sum_r |D_{wr}| \max_{r,s} |\mathcal{X}_{rs}| < 2^{63} - 1$ before every product and a separate bound before diagonal addition. It evaluates displacement bands, each having distinct row indices; the absolute row-sum bound controls every partial sum. The final squared sum, scalar coefficients, quotient and rational interval calculations use arbitrary-precision Python integers. No floating eigenvalue is a proof premise. Exact $D^t G = GD$ is checked in every case. Separate orthonormal-spin construction cross-checks the coordinate identity for $n = 2, 3$ and all three fields to floating discrepancy below 10^{-14}. This latter check is diagnostic, not a rigorous enclosure; the coordinate proof above is algebraic.

An initial SciPy import failed because that package was absent. No installation was attempted; the final program uses NumPy band products. The separate `_audit/M544-author-scalar.py` and JSON verify three boundary table transitions, the original .001 seed, both recurrence endpoint coefficients, and seven bounded levels. The infinite statement rests on the proved monotonicity induction, not those seven samples.

7.2 Independent review and its exact scope

The independent review at `_audit/M544-independent-20261007/findings.txt` and `_audit/M544-independent-20261007/results.json` reports 164 of 164 exact checks passing. It accepted coefficient six, the common-reference argument for all intermediate physical lengths, the physical Gram $G = J^{-2}$, the overflow safeguards, and the normalized filter improvement. It supplied the stronger seed (6.1) and the unchanged-schedule refinement (4.7); both are incorporated with proofs here. It audited the author's saved thermal receipts and program but did not rerun the thirty thermal recurrences. They must not be described as independently replayed twice.

Author read scope includes the entire assigned source as described in Section 1. Private corpus read ranges were complete files for `ch01d-physical-metrics.tex` and `ch02/finite.tex`, lines 1–185 of `ch01a-state-domain.tex`, and lines 286–439 of `TR-core-pat-5.tex`; all cited passages are within them. The thermal JSON records SHA-256 for every available assigned source file and these four corpus files. Hashing is not counted as reading. No private text or path was sent as a public search query.

Still unverified independently here are the companion's periodic finite spectral/variational calculations and global theorem, and the four long-chain MPS contractions. The independent reviewer additionally checked the displayed initialization scalar deductions; that does not replay their underlying physical computations. Neither a full independent source-theorem proof, historical novelty, nor an end-to-end runtime speedup is claimed. Native compiler responses, disk PDF compilation and visual QA, parent integration, and freeze hashes are separate author receipts; none is a delivery record.

8 Comparison and Research Directions

Primary public literature was checked by public topic/title on October 7, 2026. Tasaki provides the odd open-chain uniform-gap assumption and conditional topological theorem [3]. Aizenman–Nachtergaele supplies the ordered-time history framework [4], not the particular periodic numerical premises here. Matrix-product amplitudes and contractions are established [5]; normalized Poisson truncation is classical [6]; the filter minimizer is a Christoffel construction [7], specialized here to $x^2 d\nu$. These attributions concern mechanisms, not a claim that those works contain the M544 constants.

Frustration-free boundary gap criteria [8] address a different structure. The pure bilinear chain is not automatically a frustration-free projector model; deleting the AKLT biquadratic interaction does not preserve a known gap theorem. An accurate MPS energy upper bound alone is likewise not a gap lower bound.

The metric residual account exposes every rounding and truncation term and supports different integer schedules without spectral diagonalization. The finer unchanged-schedule bound may be preferable when preserving already audited quotients matters. The replacement has fewer nontrivial stages but a coarser certified error, and any runtime advantage requires measurement. The rational cubic improves the relevant normalized quantity but only slightly; it is not the main mathematical gain.

The source-specific bootstrap refinement is more consequential: it improves a uniform conditional exponent at the same field and physical length threshold, with no added trial or thermal premise. Its principal limitation is equally explicit: the upstream periodic and variational premises must all hold. The present contribution is not a new independent solution of the Haldane conjecture.

Priority future work is independent replay of those remaining inputs with exact pin and normalization matching. Further schedule work could use weighted a posteriori residuals and

invariant blocks, with actual reproducible runtime/memory comparisons. Robust polynomial optimization can incorporate full moment intervals and support localizers; higher degree requires new thermal data. Keeping separate periodic, boundary-spatial, and physical defects may improve restart constants beyond the single-common-scalar proof, but any such gain must retain all intermediate-length estimates. Field perturbations require fresh endpoint, metric, thermal, and trial checks; topological extensions require the symmetry and limit conditions of the relevant theorem.

9 Contribution Assessment

The realized internal score is C3/5. The main evidence is the proved seed-preserving induction and all-length conditional gap in Theorem 6.1 and Corollary 6.2. The thermal residual theorem, replacement schedule, thirty completed author recurrences, independent receipt/scalar review, and normalized filter certificate supply additional concrete local evidence. The tiny filter gain and nominal stage-count reduction alone would not justify a high contribution score. Periodic and long-chain variational premises prevent any claim of independently established global progress. Applicability L5 is different from this realized-contribution assessment. This is an internal evidence-based assessment, not external peer review or a historical novelty determination.

References

[1] OpenAI, *A boundary-field gap for the spin-one Heisenberg chain*, September 24, 2026, pinned study source M544. `https://github.com/openai/math/tree/adc7f1241b42e322a6451854ab7e4b4c146bf78a/preprints/A-boundary-field-gap-for-the-spin-one-Heisenberg-chain-September-24-2026`.

[2] OpenAI, *The periodic spin-one Haldane gap*, September 24, 2026, companion M543 at the same pin; its periodic estimates remain explicit premises in this report.

[3] H. Tasaki, *Ground state of the $S = 1$ antiferromagnetic Heisenberg chain is topologically nontrivial if gapped*, Physical Review Letters **134**, 076602 (2025). `https://arxiv.org/html/2407.17041v4`.

[4] M. Aizenman and B. Nachtergaele, *Geometric aspects of quantum spin states*, Communications in Mathematical Physics **164**, 17–63 (1994). `https://arxiv.org/abs/cond-mat/9310009`.

[5] D. Pérez-García, F. Verstraete, M. M. Wolf, and J. I. Cirac, *Matrix product state representations*, Quantum Information and Computation **7**, 401–430 (2007). `https://arxiv.org/abs/quant-ph/0608197`.

[6] B. L. Fox and P. W. Glynn, *Computing Poisson probabilities*, Communications of the ACM **31**, 440–445 (1988), Proposition 1. `https://web.stanford.edu/~glynn/papers/1988/FG88.pdf`.

[7] J. B. Lasserre and E. Pauwels, *The empirical Christoffel function with applications in data analysis*, Advances in Computational Mathematics **45**, 1439–1468 (2019). `https://arxiv.org/abs/1701.02886`.

[8] M. Lemm and E. Mozgunov, *Spectral gaps of frustration-free spin systems with boundary*, Journal of Mathematical Physics **60**, 051901 (2019). `https://arxiv.org/abs/1801.08915`.